# Implementation of clustering methods for the identification of cognitive diversity in two samples of children from the Metropolitan Area of Buenos Aires

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## Background

Developmental sciences recognize that individual differences in cognitive development may represent adaptations to context rather than atypical performance. Cluster analyses help identify diverse profiles, but there is a lack of studies that integrate data from different evaluations and populations, limiting their use in designing interventions.

## Objectives

A. To analyze the diversity of performance profiles in tasks with executive demands in preschool-aged children from low socioeconomic status in the Metropolitan Area of Buenos Aires.
B. To compare the implementation of clustering techniques in two samples with different demographic compositions.

## Methods

Two samples from projects developed at UNA were analyzed:
- Sample A: 58 participants (M age = 5.82; SD = 0.28)
- Sample B: 106 participants (M age = 5.34; SD = 0.33).

For both samples, inhibitory control (IC), working memory (WM), and planning (PL) processes were assessed.

Performance data were analyzed using different clustering methods (k-means, PAM, Diana, and hierarchical) and distance metrics (Euclidean and Manhattan), iterating different numbers of clusters (k) for each sample. The solutions obtained were then analyzed and compared.

All analyses were conducted using R version 4.4.2 (R Core Team, 2025) in RStudio (RStudio Team, 2024). The main packages used were dplyr (Wickham et al., 2023), ggplot2 (Wickham et al., 2025), and diceR (Chiu & Talhouk, 2025).

## Results

A final number of k = 3 clusters was selected using k-means and PAM. The groups differed significantly in all tasks ($p < .05$), showing both similarities and differences across samples.

- Cluster 1: Low performance and low reaction times (RT) in IC in both samples. In WM, low performance in Sample A and high performance in Sample B.
- Cluster 2: Medium performance in IC in both samples, with high RTs in Sample A, accompanied by low performance in PL.
- Cluster 3: Medium-high performance in IC and PL in both samples. In IC RTs, low RTs in Sample A and high RTs in Sample B.

Figure 1. Comparison of performance and reaction times of the clusters generated for each sample (z-values). *p < 0.05; **p < 0.01; ***p < 0.001.
Figure 2. Images of the neighborhoods where participants of Samples A and B lived, respectively.

## Conclusions

Clustering methods allowed the identification of heterogeneous profiles both between and within samples. These results contribute to the development of a methodological approach relevant to the design of interventions that consider children's cognitive diversity.

## References

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